

E-HANK: full model and equations

A small open economy HANK with a brown/green durable adoption margin

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Abstract

This note states the complete model implemented in the code, with all equilibrium equations. The aggregate structure is [1] (henceforth ARS): a small open economy with sticky wages, imported energy and foreign goods with sticky retail prices, and a mutual fund holding claims on domestic firms, importers and a share of the energy endowment. The only structural addition is a *discrete brown/green durable adoption choice* in the household block; every aggregate block is inherited from ARS. Sections 2–3 give the household problem and its aggregation; Sections 4–8 give firms, prices, the open economy, policy and market clearing; Section 9 gives the steady state, shocks and calibration. The balance-of-payments booking of the green margin is the one open modelling choice; it is stated in Section 6 and discussed in Section 10.

1 Conventions and notation

Time is quarterly. A subscript t is suppressed where unambiguous; $x(-1)$ and $x(+1)$ denote lag and (expected) lead. A bar denotes a steady-state value, e.g. $\bar{p}^E$. All *relative prices* carry the subscript “/ P ” where P is the CPI, e.g. $P_B^E \equiv P_{B,t}^E/P_t$; these are the objects the CES demand system uses, *regardless* of the unit of account.

Unit of account (numeraire). The household block is numeraire-generic. It reads three upstream objects,

$$p^{\text{num}} \text{ (price of the unit of account, relative to the CPI),} \quad r^{\text{num}}, \quad atw^{\text{num}}, \quad (1)$$

and returns assets denominated in units of account, converted to CPI units downstream by $A^{CPI} = Ap^{\text{num}}$. Two closures are implemented:

$$\text{cpi (ARS): } p^{\text{num}} = 1, \quad r^{\text{num}} = r, \quad atw^{\text{num}} = atw; \quad \text{core (baseline): } p^{\text{num}} = p_{H/P}, \quad 1+r^{\text{num}} = (1+r)^{\frac{p_{H/P}(-1)}{p_{H/P}}} \quad (2)$$

Under **core** the unit of account is the domestic good, so the resource constraint is independent of how the CPI weights brown and green energy. Objects contractually written in CPI units—fiscal transfers, the ARS subsistence term, and the switching cost ψ_g —are divided by p^{num} on the way into the budget. The baseline is **core**.

2 Households

There are three permanent discount-factor types $i \in \{0, 1, 2\}$ of equal mass, with β_i (Section 9.4); within a type, households face idiosyncratic labour productivity and hold a discrete durable technology.

2.1 Idiosyncratic productivity and the durable state

Labour productivity e follows a Rouwenhorst discretisation of an AR(1) in logs, $\log e' = \rho_e \log e + \varepsilon$, with n_e points, persistence ρ_e and innovation s.d. σ_e ; transition matrix Π , mean-one grid.

Each household carries two binary durable indicators: the technology it *operates this period* $g \in \{B, G\}$ and the technology it *held last period* $g_- \in \{B, G\}$. We collapse the pair into a four-point index

$$d \in \{BB, BG, GB, GG\}, \quad (\text{now, before}) = \{(B, B), (B, G), (G, B), (G, G)\}. \quad (3)$$

Define the indicators

$$\text{IS_GREEN}(d) = \mathbb{I}[g = G] = (0, 0, 1, 1), \quad \text{PAYS_SWITCH}(d) = \mathbb{I}[g = G, g_- = B] = (0, 0, 1, 0). \quad (4)$$

The switching cost ψ_g is incurred exactly in the transition $g_- = B \rightarrow g = G$ (state GB). Brown is *absorbing*; a green durable *breaks down* to brown at rate δ_g . The breakdown Markov (applied in the **dep** stage, rows = state this period, columns = state next period) is

$$M_{\text{dep}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ \delta_g & 0 & 0 & 1 - \delta_g \\ \delta_g & 0 & 0 & 1 - \delta_g \end{pmatrix}. \quad (5)$$

On the equilibrium path BG carries zero mass (no household abandons the cheaper technology; green is lost only through breakdown, recorded as $g_- = B$), so the effective state is three-point. In steady state a permanent green share is sustained by a flow of switchers,

$$D^{\text{sw}} = \delta_g D^G. \quad (6)$$

Forward stage order within a period is [**dep**, **prod**, **durables**, **consav**]: breakdown, productivity, the discrete adoption choice, and the EGM consumption–savings step. The adoption stage uses the discrete–continuous structure of [4].

2.2 Preferences and the type-specific bundle price

Flow felicity is CRRA over a consumption bundle c with disutility of labour handled at the union level (Section 4.1):

$$u(c) = \begin{cases} \log c & \text{eis} = 1 \\ \frac{c^{1-1/\text{eis}}}{1 - 1/\text{eis}} & \text{eis} \neq 1. \end{cases} \quad (7)$$

The bundle c is a CES nest of energy against a home/foreign composite, with ARS parameters (α_E, η_E) . Because a green adopter buys energy at P_G^E and a brown household at P_B^E , each durable type faces its own energy price and hence its own *exact* cost-of-living index:

$$p_d^E = P_B^E + \text{IS_GREEN}(d) (P_G^E - P_B^E), \quad (8)$$

$$p_d = \left[1 + \alpha_E ((p_d^E)^{1-\eta_E} - (P_B^E)^{1-\eta_E}) \right]^{\frac{1}{1-\eta_E}} \quad (\eta_E \neq 1), \quad p_d = (p_d^E / P_B^E)^{\alpha_E} \quad (\eta_E = 1). \quad (9)$$

Equation (9) uses the equilibrium CES identity $\alpha_E (P_B^E)^{1-\eta_E} + (1 - \alpha_E) p_{HF/P}^{1-\eta_E} = 1$ (the block **CESprices**, eq. (34)), so *brown* households have $p_d \equiv 1$ at every date (exact nesting). p_d is CPI-relative; the budget uses $p_d^{\text{num}} \equiv p_d / p^{\text{num}}$.

2.3 Cash on hand

Let a be beginning-of-period assets (in units of account, on a grid with lower bound $a \geq \underline{a}$, $\underline{a} = 0$ in the baseline). Cash on hand of a household in durable state d with productivity e is

$$\text{coh}(d, e, a) = (1 + r^{\text{num}}) a + atw^{\text{num}} e + \frac{T^f + T^{\text{fisc}} + T^{\text{green}}}{p^{\text{num}}} - \frac{\psi_g}{p^{\text{num}}} \text{PAYS_SWITCH}(d), \quad (10)$$

with the (CPI-denominated) terms

$$T^f = -P_B^E \bar{c}_E [(atw_n \mu_{\text{ss}}) s_w + (1 - s_w)] \quad \text{ARS energy subsistence } (\bar{c}_E = 0 \text{ in baseline} \Rightarrow T^f = 0), \quad (11)$$

$$T^{\text{fisc}} = \epsilon^T + \iota (p^E - \bar{p}^E) \bar{c}_{E,i}^B(e, a) \quad \text{untargeted lump sum } \epsilon^T + \text{Slutsky transfer}, \quad (12)$$

$$T^{\text{green}} = T^{\text{green}} \quad \text{domestic green-sector rebate } (= 0 \text{ under import booking}). \quad (13)$$

Here $\iota = \text{insE}$ scales the Slutsky transfer and $\bar{c}_{E,i}^B(e, a)$ is household i 's steady-state *brown* energy demand, collapsed over the durable axis and broadcast identically across durable states (so a switcher keeps the same transfer; the switching margin is undistorted). $\epsilon^T = \text{epsT}$. The switching cost enters cash on hand only in state GB .

2.4 Discrete adoption choice

In the **durables** stage the household chooses its current technology g given the technology it holds (post-breakdown), by a logit over continuation values with taste scale $\sigma_\epsilon = \text{taste_shock}$. Feasible transitions carry the flow payoff

$$f(d \mid d_-) = \begin{cases} 0 & \text{stay/renew brown: } (BB \mid BB), (BB \mid BG), \\ -\text{green_block} \quad (\text{masked to } -\infty \text{ if } \text{coh}(GB) \leq 0) & \text{switch to green: } (GB \mid BB), (GB \mid BG), \\ 0 & \text{stay/renew green: } (GG \mid GB), (GG \mid GG), \\ -\infty & \text{otherwise (infeasible)}. \end{cases} \quad (14)$$

The switching cost itself is already in $\text{coh}(GB)$ through eq. (10); **green_block** is a counterfactual wedge, 0 in the baseline (margin open) and $+\infty$ (large) in the “no-adoption” variant (margin shut). Given post-choice values $V(d)$, the choice probabilities and the expected (pre-shock) value are the standard logit / log-sum-exp:

$$\Pr(d \mid d_-) = \frac{\exp([f(d \mid d_-) + V(d)]/\sigma_\epsilon)}{\sum_{d'} \exp([f(d' \mid d_-) + V(d')]/\sigma_\epsilon)}, \quad \tilde{V}(d_-) = \sigma_\epsilon \log \sum_{d'} \exp\left(\frac{f(d' \mid d_-) + V(d')}{\sigma_\epsilon}\right). \quad (15)$$

2.5 Consumption–savings (EGM)

Conditional on the durable state, the household solves a standard consumption–savings problem with the type-specific bundle price p_d^{num} . The budget, Euler and envelope are

$$p_d^{\text{num}} c + a' = \text{coh}(d), \quad a' \geq \underline{a}, \quad (16)$$

$$u'(c) = \beta_i p_d^{\text{num}} \mathbb{E} V_a(a', e', d') \quad (\text{Euler}), \quad (17)$$

$$V_a = (1 + r^{\text{num}}) u'(c) / p_d^{\text{num}} \quad (\text{envelope}), \quad (18)$$

$$V = u(c) + \beta_i \mathbb{E} V(a', e', d') \quad (\text{value}), \quad (19)$$

solved by the endogenous grid method. With $u(c) = c^{1-1/\text{eis}}/(1 - 1/\text{eis})$, $u'(c) = c^{-1/\text{eis}}$. The consumption reported to welfare is net of the (baseline-zero) GHH term; see `welfare.py`.

2.6 Household outputs (hetoutputs)

Given the policy $c(d, e, a)$, the household's *exact* CES demand split by durable type is

$$c_E(d) = \alpha_E (p_d^E / p_d)^{-\eta_E} c, \quad c_{HF}(d) = (1 - \alpha_E) (p_{HF/P} / p_d)^{-\eta_E} c, \quad (20)$$

$$c_E^B(d) = c_E(d) (1 - \text{IS_GREEN}(d)), \quad c_E^G(d) = c_E(d) \text{IS_GREEN}(d). \quad (21)$$

(Prices in (20) are CPI-relative; the denominator is p_d , not p_d^{num} .) The block also returns $d_G(d) = \text{IS_GREEN}(d)$, $d_{sw}(d) = \text{PAYS_SWITCH}(d)$, the MPC, and flow utility $u(c)$.

3 Aggregation

For each type i , SSJ aggregates the hetoutputs over the stationary distribution Λ_i into $C_i, A_i, \text{MPC}_i, D_i^G, D_i^{\text{sw}}, C_{E,i}^B, C_{E,i}^G$, and the model averages equally across the three types:

$$X = \frac{1}{3} \sum_i X_i, \quad X \in \{C, A, \text{MPC}, D^G, D^{\text{sw}}, C_E^B, C_E^G, C_{HF}, \dots\}, \quad \beta = \frac{1}{3} \sum_i \beta_i. \quad (22)$$

Assets in units of account are converted to CPI units, $A^{CPI} = A p^{\text{num}}$. Aggregate demands for the home good, the foreign good and energy are then built from the household aggregates (block `hh_outputs_dur`), applying on top only the inner home/foreign nest, which is common to all households:

$$c_E = C_E^B + C_E^G \quad (+ \text{ARS subsistence, 0 in baseline}), \quad (23)$$

$$c_H = (1 - \alpha_F) p_{H/HF}^{-\eta} C_{HF}, \quad c_F = \alpha_F p_{F/HF}^{-\eta} C_{HF}. \quad (24)$$

Using $\sum_d c_E(d)$ rather than the CES demand at the aggregate index avoids a Jensen bias of order 0.5%.

4 Firms and domestic production

A competitive final domestic good is produced from labour (energy in production is off in the baseline, `prodE_share` = 0). With productivity Z and a constant gross markup μ_{ss} ,

$$y = Z n, \quad \text{gdp} = Z n, \quad \text{btw}_n = w n, \quad \text{atw}_n = (1 - \tau^Y) w n, \quad (25)$$

where y is home-good output, n hours, w the real wage. Monopolistic profits are capitalised into the mutual fund:

$$\text{div} = (\mu_{\text{ss}} - 1) w n, \quad \text{div}^{\text{tot}} = \text{div} + \mathbb{I}[\text{pcX_home} = 1] \text{div}^X, \quad J = \text{div}^{\text{tot}} + \frac{J(+1)}{1 + r^{\text{ante}}}, \quad j = \frac{J(+1)}{1 + r^{\text{ante}}}. \quad (26)$$

Export pricing profit under (local- vs producer-)currency pricing is $\text{div}^X = (Q^{1-\varepsilon_{\text{dcp}}} p_{H/P}^{\varepsilon_{\text{dcp}}} - p_{H/P}) c_H^*$, which is 0 in the baseline ($\varepsilon_{\text{dcp}} = 1$, producer-currency pricing).

4.1 Wage setting (sticky wages)

A union sets the nominal wage subject to Calvo frictions θ_w , giving a wage Phillips curve. With $\kappa_w = (1 - \theta_w)(1 - \beta\theta_w)/\theta_w$, labour-disutility shifter φ (**vphi**), Frisch elasticity **frisch**, and a real-wage-rigidity term controlled by w_{BG} ,

$$\pi^w(1 + \pi^w) = \beta \pi^w(+1)(1 + \pi^w(+1)) + \frac{\kappa_w}{\varphi n^{1/\text{frisch}}} \Xi, \quad (27)$$

$$\Xi = \varphi n^{1/\text{frisch}} - \frac{atw C^{-1/\text{eis}}}{\mu_{ss}} - w_{BG} \underbrace{\frac{(w - \bar{w}) w \bar{C}^{-1/\text{eis}} \bar{n}}{\mu_{ss} n \bar{w}}}_{\text{real-wage rigidity}} + \text{wedge}, \quad (28)$$

and the nominal wage and real wage evolve as $W = (1 + \pi^w)W(-1)$, $w = W/P$. Here $atw = atw_n/n$ is the after-tax wage per hour.

5 Prices

All retail prices are relative to the CPI P ; Q is the real exchange rate (price of the foreign basket in home units), P_F^*, P^{E*} are exogenous world prices.

Domestic good. With `prodE_share` = 0,

$$P_H = \frac{\mu_{ss}}{Z} W, \quad p_{H/P} = P_H/P, \quad \pi^H = P_H/P_H(-1) - 1, \quad w = W/P. \quad (29)$$

Imported foreign good (Calvo θ_F). With $\kappa_F = (1 - \theta_F)(1 - \beta_F\theta_F)/\theta_F$, $\beta_F = 1/(1 + \bar{r}^{\text{ante}})$,

$$\kappa_F \left(\frac{QP_F^*}{p_{F/P}} - 1 \right) + \beta_F \pi^F(+1)(1 + \pi^F(+1)) - \pi^F(1 + \pi^F) = 0, \quad p_{F/P} = P_F/P. \quad (30)$$

Energy retail price and the price cap (Calvo θ_E). The wholesale/retail brown energy price $p^E = P_E/P$ satisfies a NKPC, and the *household* brown price applies the cap:

$$\kappa_E \left(\frac{QP^{E*}}{p^E} - 1 \right) + \beta_E \pi^E(+1)(1 + \pi^E(+1)) - \pi^E(1 + \pi^E) = 0, \quad \kappa_E = (1 - \theta_E)(1 - \beta_E\theta_E)/\theta_E, \quad (31)$$

$$P_B^E = (1 - \tau^E) p^E + \tau^E \bar{p}^E, \quad \tau^E = 1 \Rightarrow P_B^E \text{ pinned at } \bar{p}^E \quad (32)$$

Green energy price. With green/brown ratio $\varrho = \text{pE_g_ratio}$ and pass-through $\rho_g = \text{pass_g}$,

$$P_G^E = \varrho \bar{p}^E (P_B^E/\bar{P}_B^E)^{\rho_g}. \quad \rho_g = 0 \text{ (baseline)} \Rightarrow P_G^E \text{ constant}. \quad (33)$$

CPI (brown-anchored; “Option C”). The energy leg of the CPI is anchored on the brown price alone. With $\alpha = \alpha_E + (1 - \alpha_E)\alpha_F$, the inner (energy vs home/foreign) and outer (home vs foreign) CES identities that define $p_{HF/P}$ and $p_{H/HF}$ are

$$(1 - \alpha_E) p_{HF/P}^{1-\eta_E} + \alpha_E (P_B^E)^{1-\eta_E} = 1, \quad (34)$$

$$(1 - \alpha_F) p_{H/HF}^{1-\eta} + \alpha_F p_{F/HF}^{1-\eta} = 1, \quad (35)$$

with $p_{F/HF} = p_{F/P}/p_{HF/P}$, $p_{H/P} = p_{H/HF} p_{HF/P}$, $p_H^* = p_{H/P}/Q$. The gap between this measured CPI and the share-weighted (“true”) deflator is computed *ex post*, outside the model:

$$\phi^{1-\eta_E} = 1 + \alpha_E D^G [(P_G^E)^{1-\eta_E} - (P_B^E)^{1-\eta_E}]. \quad (36)$$

Under the domestic-good numeraire this choice affects only the measured deflator and the monetary rule, not the resource constraint.

Aggregate price level and inflation.

$$P = (1 + \pi)P(-1), \quad \pi^{\text{ann}} = (1 + \pi)^4 - 1, \text{ etc.} \quad (37)$$

6 Open economy

Foreign demand for the home good. With foreign demand shifter α^* , trade elasticity γ , and world demand C^* ,

$$c_H^* = \alpha^* (p_H^*)^{-\gamma \varepsilon_{dcp}} C^*. \quad (38)$$

Uncovered interest parity. The real exchange rate solves

$$\frac{Q}{Q(+1)} (1 + r^{\text{ante}}) = 1 + r^*. \quad (39)$$

Importer profit centres. Retail importers of the foreign good and of energy earn margins over the world price, capitalised into the fund:

$$D_F = (p_{F/P} - Q P_F^*) c_F, \quad J_F = D_F + \frac{J_F(+1)}{1 + r^{\text{ante}}}, \quad j_F = \frac{J_F(+1)}{1 + r^{\text{ante}}}, \quad (40)$$

$$D_E = (p^E - Q P^{E*}) (c_E + \text{prodE}), \quad J_E = D_E + \frac{J_E(+1)}{1 + r^{\text{ante}}}, \quad j_E = \frac{J_E(+1)}{1 + r^{\text{ante}}}. \quad (41)$$

(Under the domestic booking the energy importer margin is levied on brown energy C_E^B only.)

World energy supply (IEA block). World supply is a fixed quantity E^{shock} plus intertemporal stock arbitrage with cost Γ_{arb} ; its rents are capitalised, and the home economy owns a share ζ^E :

$$E_{\text{stock}} = \frac{P^{E*} (+1) / (1 + r^*) - P^{E*}}{\Gamma_{\text{arb}}}, \quad E^{\text{supply}} = E^{\text{shock}} + (E_{\text{stock}}(-1) - E_{\text{stock}}), \quad (42)$$

$$D_{E^*} = Q P^{E*} E^{\text{supply}}, \quad J_{E^*} = D_{E^*} + \frac{J_{E^*} (+1)}{1 + r^{\text{ante}}}, \quad j_{E^*} = \frac{J_{E^*} (+1)}{1 + r^{\text{ante}}}. \quad (43)$$

Fund return (revaluation). The realised (ex post) return on the fund’s claims defines r :

$$1 + r = \frac{J + \zeta_F J_F + \zeta_E J_E + \zeta^E J_{E^*}}{j(-1) + \zeta_F j_F(-1) + \zeta_E j_E(-1) + \zeta^E j_{E^*}(-1)}, \quad (\zeta_F = \zeta_E = 0 \text{ baseline}), \quad (44)$$

and analogously the domestic-fund return r^{dom} and value A^{dom} (used only in the current-account revaluation term).

Balance of payments (import booking, baseline). Let the two durable-margin terms be

$$\text{imports}_{\text{dur}} = \psi_g D^{\text{sw}} \quad (\text{switching cost booked as import}), \quad \text{energy_gap} = (P_B^E - P_G^E) C_E^G \quad (\text{adopters' import saving}) \quad (45)$$

Exports, imports and net exports are

$$\text{exports} = Q(p_H^* c_H^* + P^{E*} \zeta^E E^{\text{supply}}), \quad (46)$$

$$\text{imports} = Q(P_F^* c_F + P^{E*} (c_E + \text{prodE})) + \text{imports}_{\text{dur}} - \text{energy_gap}, \quad (47)$$

$$nx = Q(p_H^* c_H^* - P_F^* c_F - P^{E*} (c_E + \text{prodE} - \zeta^E E^{\text{supply}})) - \text{imports}_{\text{dur}} + \text{energy_gap}. \quad (48)$$

Net foreign assets accumulate with a revaluation term:

$$\text{reval} = (r - r^{\text{ante}}(-1)) A^{CPI}(-1) - (r^{\text{dom}} - r^{\text{ante}}(-1)) A^{\text{dom}}(-1), \quad (49)$$

$$\text{nfa} = nx + \text{reval} + (1 + r^{\text{ante}}(-1)) \text{nfa}(-1), \quad nx/\text{gdp} = nx/y. \quad (50)$$

7 Policy

7.1 Monetary policy

The nominal rate follows an inertial rule; the ex-ante real rate is Fisher:

$$1 + i = \rho_i (1 + i(-1)) + (1 - \rho_i) (1 + \bar{r}^*) (1 + \phi_\pi \pi) (1 + \phi_{\pi w} \pi^w) (1 + \phi_{\pi e} \pi(+1)) + \text{ishock}, \quad (51)$$

$$r^{\text{ante}} = \frac{1 + i}{1 + \pi(+1)} - 1. \quad (52)$$

The baseline is the ARS *constant real rate* rule $(\rho_i, \phi_\pi, \phi_{\pi w}, \phi_{\pi e}) = (0, 0, 0, 1) \Rightarrow r^{\text{ante}} = \bar{r}^*$; a standard Taylor rule sets $\phi_\pi = 1.5, \phi_{\pi e} = 0$. **ishock** is a monetary shock.

7.2 Fiscal policy

The government runs the two energy-crisis instruments and a lump sum, all debt financed, with debt stabilised through the distortionary labour tax. Define the fiscal base of the subsidy (booking-dependent):

$$\text{base}_S = \begin{cases} c_E & \text{import booking} \\ C_E^B & \text{domestic booking.} \end{cases} \quad (53)$$

Then

$$\text{Subsidy} = \tau^E (p^E - \bar{p}^E) \text{base}_S \quad \text{price cap (cost scales with } p^E) \quad (54)$$

$$\text{Ttargeted} = \iota (p^E - \bar{p}^E) \bar{C}_E^B \quad \text{Slutsky transfer (cost scales with } \bar{C}_E^B) \quad (55)$$

$$\text{Tuntargeted} = \epsilon^T, \quad (56)$$

$$\text{spending} = \text{Tuntargeted} + \text{Ttargeted} + \text{Subsidy}, \quad \text{taxation} = \tau^Y \text{btw}_n = \tau^Y w n. \quad (57)$$

The government budget constraint and the debt-stabilising tax rule are

$$B = (1 + r^{\text{ante}}(-1)) B(-1) + \text{spending} - \text{taxation}, \quad (58)$$

$$\tau^Y = (1 - \text{bb}) \psi_B (B(-1) - \bar{B}) + \text{bb} (B - \bar{B}), \quad (59)$$

with feedback coefficient $\psi_B = \text{psiB}$. **In the baseline $\bar{B} = 0$ (no steady-state government debt) and $\text{bb} = 0$, so (59) reduces to $\tau^Y = \psi_B(B(-1) - \bar{B})$:** crisis outlays are deficit-financed and repaid by raising the labour tax in proportion to the debt's deviation from zero. Government debt B is a genuine transition unknown (it moves in the dynamics) and enters household wealth through asset market clearing, eq. (61).

8 Market clearing and equilibrium

The equilibrium residuals are

$$\text{goods: } c_H + c_H^* - y = 0, \quad (60)$$

$$\text{assets (Walras): } A^{CPI} - \text{nfa} - j - j_F - j_E - B - \zeta^E j_{E^*} = 0, \quad (61)$$

$$\text{energy: } \begin{cases} P^{E^*} - P^{E^*, \text{shock}} = 0 & \varepsilon^E = \infty \text{ (price taker; price shock),} \\ (c_E + \text{prodE}) - E^{\text{supply}} = 0 & \varepsilon^E < \infty \text{ (fixed quantity; supply shock).} \end{cases} \quad (62)$$

Under the domestic booking, only brown energy clears against world supply in (62): $(C_E^B + \text{prodE}) - E^{\text{supply}} = 0$. Equation (61) is a Walras'-law residual (not itself a solved target); it is monitored to machine precision as a consistency check.

Transition unknowns/targets. The linear (SSJ) transition solves for $\{y, \pi, r, \tau^Y, P^{E^*}, w\}$ against $\{\text{goods}, \pi \text{ residual}, r \text{ residual}, \tau^Y \text{ residual}, \text{energy}, w \text{ residual}\}$; under the domestic booking T^{green} is an added unknown with residual (65).

9 Steady state, shocks, calibration

9.1 Openness and endowment normalisations

The openness parameters and the ARS home-endowment normalisation ($z \equiv \zeta^E$, $a_E \equiv \alpha_E$) are

$$\alpha_F = \frac{\alpha - \alpha_E}{1 - \alpha_E}, \quad \gamma = \frac{\chi_{\text{tgt}} - (1 - \alpha)(1 - \alpha_F)\eta_E}{(1 - \alpha)\alpha_F + \alpha}, \quad \eta = \gamma, \quad \chi_{\text{tgt}} = 0.3, \quad (63)$$

$$y = 1 - z a_E, \quad \mu_{\text{ss}} \leftarrow \mu_{\text{ss}}(1 - z a_E), \quad \alpha^* = \alpha - a_E z, \quad Z = \mu_{\text{ss}}, \quad E^{\text{shock}} = a_E. \quad (64)$$

So the gross markup entering firm profits is $\mu_{\text{ss}}(1 - \zeta^E \alpha_E) \approx 1.016$; 1.03 is the pre-normalisation primitive.

9.2 Steady-state unknowns/targets

The steady state solves for $\{\varphi, \beta_{\text{max}}, y, \psi_g\}$ against $\{\pi^w \text{ res}, \text{nfa res}, \text{goods}, D^G - 0.05\}$, i.e. ψ_g is calibrated to the level target $D_{\text{ss}}^G = 0.05$ and β_{max} to close net foreign assets. Under the domestic booking T^{green} is an added unknown closed by the zero-profit green-sector residual

$$T^{\text{green}} = P_G^E C_E^G + \psi_g D^{\text{sw}}. \quad (65)$$

In the “no-adoption” counterfactual (**green.block** large) $D^G \equiv 0$ for every ψ_g , so ψ_g is not identifiable and is carried over, fixed, from the matching adoption steady state.

9.3 Shocks

Brown-price shock (requires $\varepsilon^E = \infty$): $P_t^{E*,\text{shock}} = \text{size} \cdot \rho^t$, $\rho = 2^{-1/\text{half-life}}$, (66)

Brown-supply shock (requires $\varepsilon^E < \infty$): $E_t^{\text{shock}} = \bar{E}^{\text{shock}} - \text{drop} \cdot \bar{E}^{\text{shock}}$ for $t < \text{quarters}$, else $\bar{E}^{\text{shock}}$. (67)

The baseline price shock is $\text{size} = 1$ (world price doubles on impact), half-life 16; the baseline supply shock is -10% for 6 quarters.

9.4 Calibration

Parameter	Value	Source / target
α_E energy expenditure share	0.04	ARS
η_E energy substitution	0.10	ARS
α total import share	0.30	ARS
r, μ_{ss} (primitive)	0.01, 1.03	ARS
eis, Frisch	1, 0.5	ARS
ρ_e, σ_e, n_e	0.92, 0.57, 7	ARS
β types (3), spread	0.06	implied MPC ≈ 0.30
$\theta_w, \theta_E, \theta_F$ (Calvo)	0.938, 0.65, 0.90	ARS
w_{BG} real-wage rigidity	5	ARS
ζ^E home energy-rent share	0.33	ARS
δ_g green breakdown rate	0.05	assumption (≈ 5 -yr life)
$\varrho = P_G^E/P_B^E$ steady-state gap	0.80	heat-pump / EV delivered-cost ratio
ψ_g switching cost	0.538	solved to $D_{\text{ss}}^G = 0.05$
σ_ε logit taste scale	0.05	calibrated jointly with ψ_g
ρ_g pass-through to green price	0	upper bound on the channel
$\tau^E, \iota, \epsilon^T$ policy	0	laissez-faire baseline
B steady-state debt	0	no SS government debt
ψ_B debt-feedback in τ^Y	0.04	

Table 1: Baseline calibration. ψ_g and σ_ε are jointly calibrated to the *level* target $D_{\text{ss}}^G = 0.05$ and are not separately identified by an *adoption-elasticity* moment (an open item).

10 Remark on the balance-of-payments booking

The one open modelling choice is how the green margin enters the balance of payments. The code implements two internally consistent bookings:

- (i) **Import (baseline above).** The switching cost $\psi_g D^{\text{sw}}$ is booked as an import and the adopter's energy saving $(P_B^E - P_G^E)C_E^G$ as an import reduction; all energy is imported at the world brown price (Section 6).
- (ii) **Domestic.** Green energy $P_G^E C_E^G$ and installation $\psi_g D^{\text{sw}}$ are domestic value added rebated lump-sum (T^{green} , eq. (65)); only brown energy is imported and clears against world supply. It is an *upper bound* on domestic value added (zero marginal cost).

The choice has a first-order effect on the adoption channel’s *output* sign. Per the current plan we intend to fix a *single* baseline: adoption purchases (the switching cost) expensed as the *core consumption good* (the home/foreign composite, split by α), and *green energy imported* on the same footing as brown. This is a third, intermediate convention—between (i) and (ii)—and is the most standard treatment (a durable installation is an ordinary consumption-good purchase). It removes the auxiliary channel that distinguishes brown from green energy at the balance-of-payments level, at the cost of re-specifying `hh_outputs_dur` (route $\psi_g D^{\text{sw}}$ through C_{HF}), the current account, and the energy importer margin.

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